AlcorCut Pro — Gimbal support: permissions and licences

Notes on adding control of DJI Osmo Mobile and Zhiyun Smooth gimbals to the Capture app. Status: **future idea, not on the active to-do list** (see FUTURE_IDEAS.md).

What works already — no permission needed

- Many phone gimbals have a shutter / record button that acts like a Bluetooth keyboard and sends a **volume key** press.
- AlcorCut Pro's **Settings** → **Side buttons** (volume key = shutter / record) already handles that, so starting and stopping recording from the gimbal handle usually works today.
- Only **motor control** from the app (pan, tilt, follow / tracking, zoom wheel) needs the gimbal maker's developer kit — or the route on page 2.

DJI Osmo Mobile

- DJI has **no public developer kit for Osmo Mobile gimbals on Android**. Its Mobile SDK covers drones and some other products; a DJI developer-forum thread says the Osmo gimbals aren't supported.
- Apps like Filmic Pro got Osmo support through **direct partner agreements** with DJI. This is not something you can buy or license off the shelf — you would apply to DJI as a partner, and DJI decides.
- The newest Osmo Mobile 8 works with Apple's own gimbal system (DockKit) on iPhone — that doesn't exist on Android.

Zhiyun Smooth

- Zhiyun has, as far as is known, offered a developer kit to app makers **on request**.
- That usually means signing their developer agreement rather than paying a fee — but the current terms need to be checked with Zhiyun before building anything.

For a paid app

- Any gimbal kit must allow use in a **paid app** with credits, like the other parts of AlcorCut Pro.
- Gimbal maker names (DJI, Osmo, Zhiyun, Smooth) belong to their owners and may only be used to describe compatibility.
- Get the maker's written OK (partner agreement or SDK licence) before shipping gimbal control.

If you decide to go ahead

- Contact Zhiyun's developer team for their SDK and terms (likely the easiest route).
- Apply to DJI's partner programme for Osmo Mobile access (uncertain; DJI chooses its partners).
- Once a kit and its licence are approved, the app side is: connect over Bluetooth, map pan / tilt / follow / zoom to the app, and add the gimbal's buttons to the Fn button choices.
- Without DJI's OK: see **Option B** on the next page.

Option B — DJI Osmo without DJI: own code from the Bluetooth protocol

Writing our own code that talks to the gimbal over Bluetooth, worked out by watching how the gimbal and DJI's own app talk to each other. Possible, but it carries legal and practical risk, so **a lawyer should check it before anything ships**.

Why it can be legal

- **Copyright protects code, not how a device communicates.** A Bluetooth command format is a way of working, not DJI's code — writing our own code that sends the same commands copies nothing of theirs.
- **EU law allows reverse engineering for interoperability** (Software Directive 2009/24/EC, article 6): studying a program to make your own independent program work with it is allowed when the information isn't otherwise available, and only the parts needed are used. Article 8 makes contract terms that forbid this void.
- **Watching the Bluetooth traffic** between the gimbal and the DJI Mimo app (e.g. Android's built-in "Bluetooth HCI snoop log") is the cleanest method — it doesn't involve taking DJI's app apart.
- The **Faroe Islands** are not in the EU. Faroese copyright law is based on the Danish one, but whether the same interoperability rule applies there — and in each country where the app is sold — must be confirmed.

Where the risk is

- **DJI's terms of use** for Mimo and its firmware forbid reverse engineering. In the EU that can't override article 6, but it can still lead to a dispute — and in other countries it may hold.
- **Locks and encryption:** if the gimbal requires a secret key or encrypted handshake, getting around it can count as breaking a "technological protection measure" (EU InfoSoc Directive, article 6; in the USA DMCA §1201, which has a narrower interoperability exception). That is where the legal risk is highest: if a lock has to be broken, stop and get legal advice first.
- **Trademarks:** DJI, Osmo and Mimo may only describe compatibility ("works with DJI Osmo Mobile 6"), never as if DJI made or approved the feature. No DJI logos.
- **Practical risk:** DJI can change the protocol with a firmware update and the feature stops working; DJI or Google Play could also receive a complaint. Not a reason not to do it, but a reason to keep it optional.

How to do it safely, if you go ahead

- **Lawyer first:** a short review by an IP lawyer (Denmark / Faroe Islands) of the plan below, before work starts and again before release.
- **Only own code:** no DJI code, libraries, keys, firmware or files copied into the app — the same rule as the rest of AlcorCut Pro.
- **Clean-room method:** one step writes down the protocol as observed (commands, values), the code is written only from those notes.
- **No circumvention:** if the protocol turns out to need a DJI key or encryption to be broken, the feature is dropped.
- **Clear label:** "Unofficial — not made or endorsed by DJI" in the app and on the store page; compatibility named per model.
- **Optional and easy to remove:** off by default, in its own part of the code, so it can be switched off remotely in an update if needed.
- Open-source projects that have mapped these protocols must still have a licence that allows a paid app with credits before anything is used from them — otherwise only our own notes and code.

Summary

- Zhiyun: ask for the SDK — lowest risk.
- DJI with permission: partner programme — no legal risk, but DJI decides.
- DJI without permission: own code from the observed Bluetooth protocol — possible within EU interoperability rules if no lock is broken, with lawyer review first and a clear "unofficial" label.

Not legal advice: these notes describe the general rules; the exact position depends on the country and on what the gimbal's protocol turns out to need. Sources: DJI SDK forum (Osmo gimbal support), DJI Mobile SDK introduction (developer.dji.com), Apple developer forum (Osmo Mobile 8 DockKit), Directive 2009/24/EC arts. 6 and 8, Directive 2001/29/EC art. 6, 17 U.S.C. §1201(f).